Exp. No: 2 Determination and plotting of Electric field and Electric flux density in sphere full of charges with varying radius.

Aim:

To write program for electric field and electric flux density in sphere using MATLAB**.**

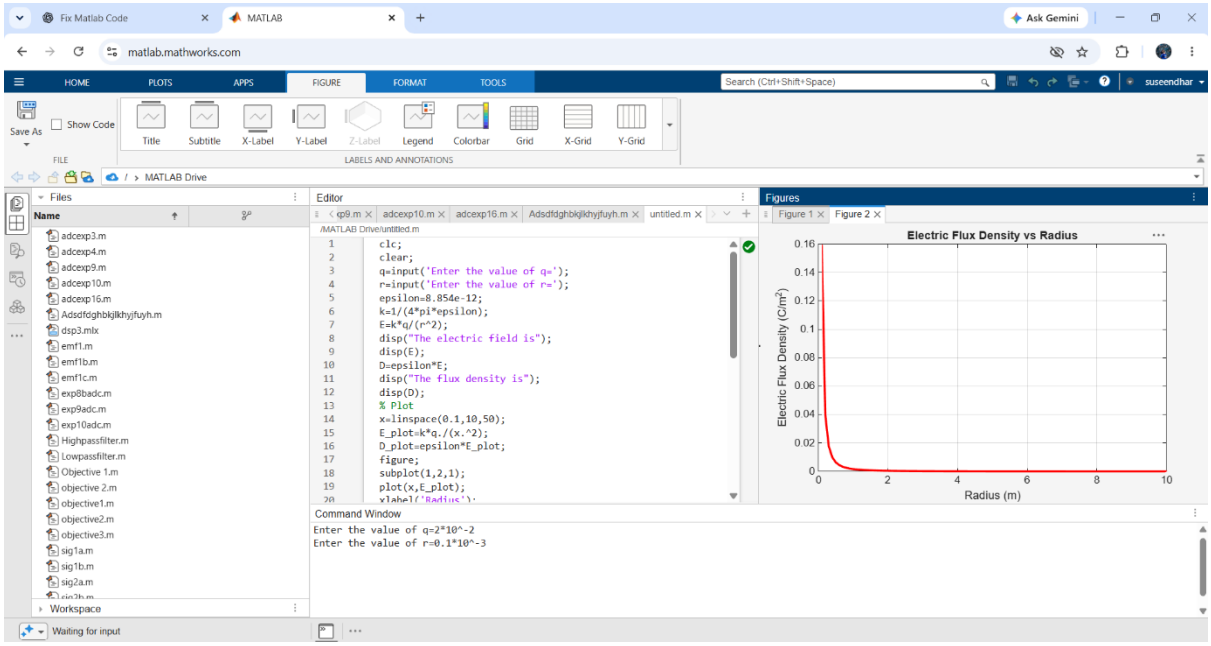
Program

(i) Determination and plotting of electric field and electric flux density in sphere full of charges with varying radius

```
clc;
clear;
q=input('Enter the value of q=');
r=input('Enter the value of r=');
epsilon=8.854e-12;
k=1/(4*pi*epsilon);
E=k*q/(r^2);
disp("The electric field is");
disp(E);
D=epsilon*E;
disp("The flux density is");
disp(D);
% Plot
x=linspace(0.1,10,50);
E_plot=k*q./(x.^2);
D_plot=epsilon*E_plot;
figure;
```

```
subplot(1,2,1);
plot(x,E_plot);
xlabel('Radius');
ylabel('Electric field');
subplot(1,2,2);
plot(x,D_plot);
xlabel('Radius');
ylabel('Flux density')
```

OUTPUT

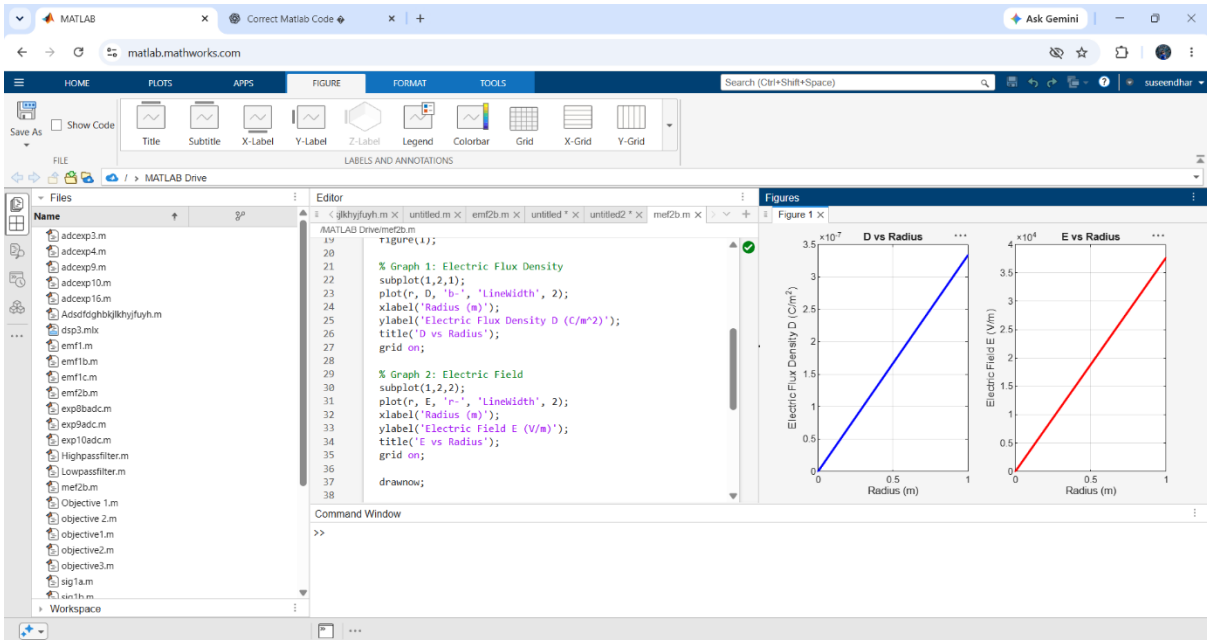


(ii) Derive and plot the electric field (E) and electric flux density (D) inside a uniformly charged sphere with charge density ρ0

```
clc;
rho0 = 1e-6;
epsilon = 8.854e-12;
R = 1;
```

```
r = linspace(0, R, 100);
D = (rho0 .* r) / 3;
E = D / epsilon;
subplot(1,2,1);
plot(r,D,'LineWidth',2);
xlabel("Radius (m)");
ylabel("Electric Flux Density (D)");
title("Uniform Density: D vs Radius");
subplot(1,2,2);
plot(r,E,'LineWidth',2);
xlabel("Radius (m)");
ylabel("Electric Field (E)");
title("Uniform Density: E vs Radius");
```

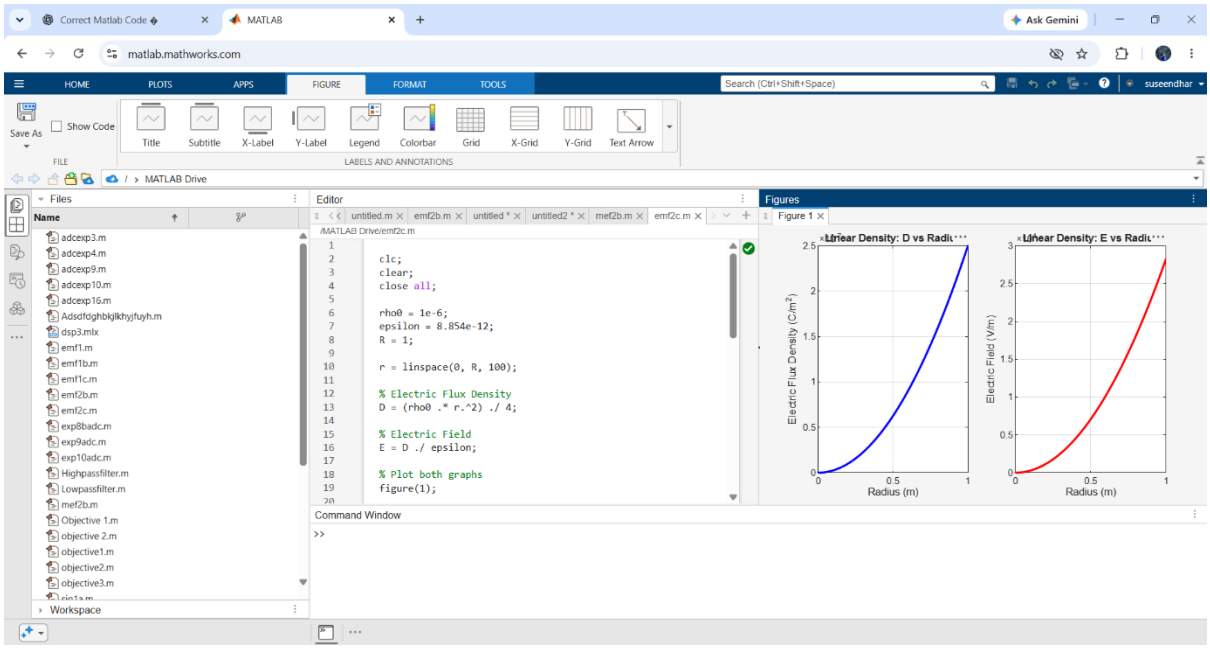
OUTPUT



(iii) For a sphere with a linearly increasing charge density ρv = ρ0r, derive and plot the electric field (E) and electric flux density (D)

```
clc;
rho0 = 1e-6;
epsilon = 8.854e-12;
R = 1;
r = linspace(0, R, 100);
D = (rho0 .* r.^2) / 4;
E = D / epsilon;
subplot(1,2,1);
plot(r,D,'LineWidth',2);
xlabel("Radius (m)");
ylabel("Electric Flux Density (D)");
title("Linear Density: D vs Radius");
subplot(1,2,2);
plot(r,E,'LineWidth',2);
xlabel("Radius (m)");
ylabel("Electric Field (E)");
title("Linear Density: E vs Radius");
```

OUTPUT



(iv) A sphere of radius R = 2 m has a uniform charge density ρ = 3×10−6 C/m3. Calculate and plot electric field inside and outside the sphere

clc;

rho0 = 3e-6;

epsilon = 8.854e-12;

R = 2;

r = linspace(0.1, 4, 100);

E = zeros(size(r));

for i = 1:length(r)

if r(i) < R

E(i) = rho0 * r(i) / (3 * epsilon);

else

Q = rho0 * (4/3) * pi * R^3;

k = 1/(4*pi*epsilon);

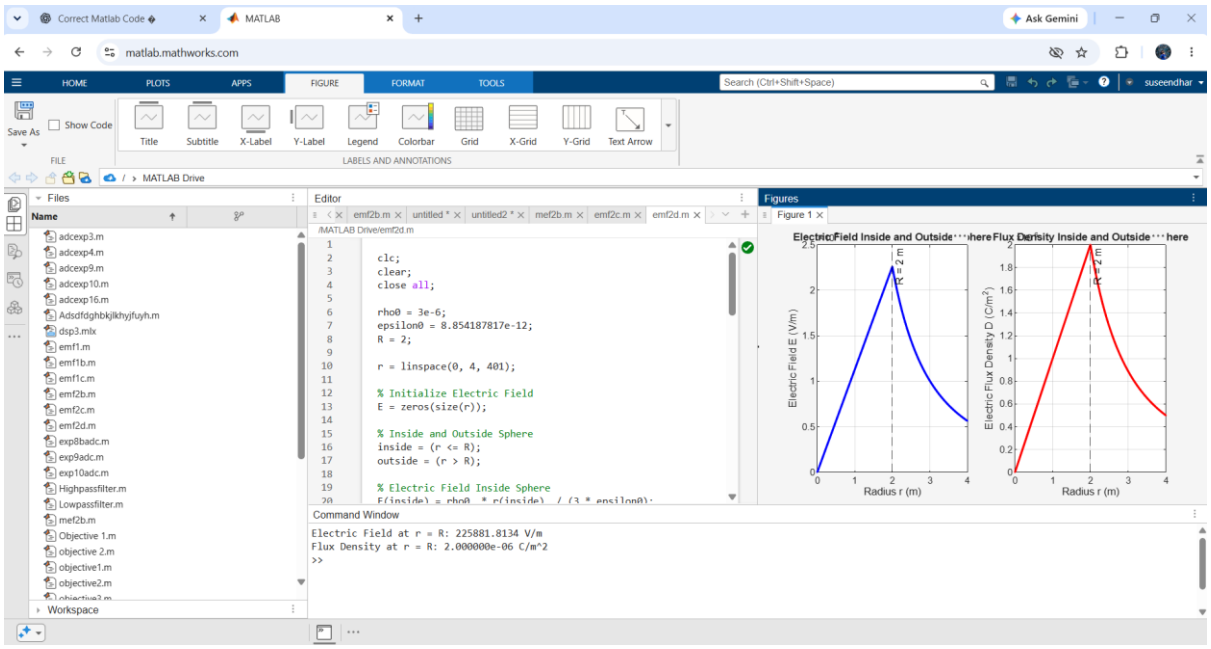
E(i) = k * Q / (r(i)^2);

end

end

```
plot(r,E,'LineWidth',2);
xlabel("Radius (m)");
ylabel("Electric Field (E)");
title("Electric Field Inside and Outside Sphere");
```

OUPUT



S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained
1	Select the appropriat e chargedistribution model	Select the appropriate charge-distribution model	Correctly distinguishes point, uniform-volume and specified non-uniform charge models and their assumptions.	Selects the correct model with minor clarification needed.	Shows partial understanding of the charge model.	Uses an unsuitable model or assumptions.	25	2.5	
2	Implement E and D calculations in MATLAB	Implement E and D calculations in MATLAB	Correctly calculates electric field E and flux density D with consistent SI units.	Calculations are mostly correct with minor coding or unit errors.	Some calculations work but require assistance.	Unable to obtain valid E and D values.	25	2.5	
3	Analyze radial variation and	Analyze radial variation and	Plots the correct inside/outside or	Plots and validation are mostly correct with minor	Plots or theoretical validation	Plots are incorrect or results	30	3	

	validate results	**validate results**	**stated distribution behaviour and verifies against theory.**	**discrepancies.**	**are incomplete.**	**are not validated.**			
4	**Interpret and document findings**	**Interpret and document findings**	**Explains the effect of radius and charge distribution and presents labelled plots/tables.**	**Interpretation is generally correct with minor omissions.**	**Interpretation is limited or documentation incomplete.**	**Cannot interpret or document results.**	**20**	**2**	
Total							**100**	**10**	